

Q.1) Display all the files from current directory which are created in particular month

[10 Marks]

```
#include <stdio.h>

#include <stdlib.h>

#include <dirent.h>

#include <sys/stat.h>

#include <time.h>


int main() {

    DIR *dir;

    struct dirent *entry;

    struct stat st;

    int month;


    printf("Enter month number (1-12): ");

    scanf("%d", &month);


    if (month < 1 || month > 12) {

        printf("Invalid month.\n");

        return 1;

    }


    dir = opendir(".");

    if (!dir) {

        perror("opendir");

        return 1;

    }


    printf("Files created/modified in month %d:\n", month);

    while ((entry = readdir(dir)) != NULL) {

        if (stat(entry->d_name, &st) == -1) {

            perror(entry->d_name);

            continue;

        }

    }

}
```

```

    }

    struct tm *tm_info = localtime(&st.st_mtime);
    if ((tm_info->tm_mon + 1) == month) { // tm_mon: 0-11
        printf("%s\n", entry->d_name);
    }
}

closedir(dir);
return 0;
}

```

Q.2) Write a C program which create a child process which catch a signal sighup, sigint and sigquit.

The Parent process send a sighup or sigint signal after every 3 seconds, at the end of 30 second parent send sigquit signal to child and child terminates my displaying message "My DADDY has Killed me!!!". [20 Marks]

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

void handle_signal(int sig) {
    if (sig == SIGHUP) {
        printf("Child received SIGHUP\n");
    } else if (sig == SIGINT) {
        printf("Child received SIGINT\n");
    } else if (sig == SIGQUIT) {
        printf("My DADDY has Killed me!!!\n");
        exit(0);
    }
}

int main() {

```

```

pid_t pid = fork();

if (pid < 0) {
    perror("fork");
    exit(1);
}

if (pid == 0) {
    // Child process: setup signal handlers
    signal(SIGHUP, handle_signal);
    signal(SIGINT, handle_signal);
    signal(SIGQUIT, handle_signal);

    while (1) {
        pause(); // Wait for signals
    }
} else {
    // Parent process
    int count = 0;
    while (count < 10) { // 10 * 3 sec = 30 sec
        sleep(3);
        if (count % 2 == 0) {
            kill(pid, SIGHUP);
        } else {
            kill(pid, SIGINT);
        }
        count++;
    }

    // After 30 seconds, terminate child
    kill(pid, SIGQUIT);
    wait(NULL);
    printf("Parent process exiting.\n");
}

```

```
    return 0;  
}
```